

Education

The Chinese University of Hong Kong Ph.D., Information Engineering. Advisor: [Angela Yingjun Zhang](#). *Hong Kong. 2026~present*

Research Areas My research primarily studies lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted. My recent work aims to improve the reliability, interpretability, and controllability of AI models in heterogeneous, computation-constrained, and communication-constrained environments.

Zhejiang University M.Sc. in Artificial Intelligence. Advised by [Meng Zhang](#).

Zhejiang. Sep 2022 ~ Dec 2025

Shandong University B.Eng. in Communication Engineering.

Shandong. Sep 2018 ~ Jul 2022

Experience

Doctoral Research @ The Chinese University of Hong Kong

Hong Kong. 2026 ~ present

Investigate data lifecycle management in distributed AI systems, connecting data generation, use, and deletion with federated representation geometry.

- Proposed an optimal-transport view of federated learning and a barycentric multi-prototype classifier for communication-constrained representation geometry. Paper #5

Research on Data-Centric ML Systems @ Zhejiang University

Zhejiang. Mar 2023 ~ Dec 2025

Built unlearning methods for data deletion across continuous influence weights, certified online updates, and dynamic tree ensembles.

- Introduced soft-weighted unlearning for continuous influence weights, supporting fairness and robustness interventions beyond binary erasure. Paper #2
- Developed Hessian-free online certified unlearning with recollected trajectory statistics, avoiding explicit Hessian inversion for streaming deletion requests. Paper #3
- Built an exact and efficient random-forest unlearning framework for dynamic online environments, updating affected tree statistics instead of retraining. Paper #4

Research on Trustworthy LLM Systems @ NUSRI CQ

Chongqing. Jun 2025 ~ Dec 2025

Analyzed reliability failures that emerge when models train on recursively selected synthetic data or infer spurious patterns from prompts.

- Showed that sample selection during recursive synthetic-data training can prune tail samples under low-resource verification and precipitate model collapse. Paper #1
- Analyzed illusory pattern perception as a mechanism for spurious LLM inference when perceived patterns override evidence-grounded reasoning. Paper #6

Open-Source Contributions and Services

🔗 **Research code releases**, public code for certified unlearning, soft-weighted unlearning, and model-collapse work.

🏠 **Xinbaopedia**, public academic homepage and wiki-style research archive with papers, figures, CV, and project notes.

📄 **Peer-Reviewing**, reviewer for ICML 2026, NeurIPS 2025 and 2026, ICLR 2025, AAAI 2025 and 2026, and IEEE TNNLS.

Selected Publications

Asterisks (*) denote co-first authorship. Accepted papers are listed before under-review manuscripts.

1. When **Sample Selection Bias** Precipitates **Model Collapse**

 ICML 2026.

Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang

Low-resource verification can turn local sample selection into persistent tail pruning.

2. Beyond Binary Erasure: **Soft-Weighted Unlearning** for Fairness and Robustness

 AAAI 2026.

Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang

Soft-weighted corrective unlearning for non-binary fairness and robustness interventions.

3. **Hessian-Free** Online Certified Unlearning

 ICLR 2025.

Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei

Certified unlearning without explicit Hessian inversion through recollected statistics.

4. **DynFrs**: An Efficient Framework for Machine Unlearning in Random Forest

 ICLR 2025.

Shurong Wang, Zhuoyang Shen, Xinbao Qiao, Tongning Zhang, Meng Zhang

Exact and efficient random-forest unlearning in dynamic environments.

5. **Federated Learning as Optimal Transport**: Barycentric Multi-Prototype Classification

Under review.

Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang

Barycentric multi-prototype classification for federated representation geometry.

6. **Illusory Pattern Perception** Drives Spurious Inference in Large Language Models

Under review.

Peihua Mai, Zhuoyan Shao, Xinbao Qiao, Meng Zhang, Xinyue Zhou, Yan Pang

Analysis of spurious pattern perception and interpretable behavior in LLMs.